

Education

University of San Francisco
Bachelor of Science - Physics

Graduated August 2019
Minors in Engineering & Computer Science

Experience

Aqua Finance, Inc. - Remote

February 2026 - Present

Manager of Analytics, Counterparty Risk

- Built counterparty risk frameworks, KPIs, early-warning indicators, predictive models, and dealer scorecards to monitor credit/fraud risk across dealers, applications, and sales channels
- Implemented monitoring rules, alerts, dashboards, and risk committee materials communicating portfolio health, dealer trends, and fraud insights to leadership

Forbright Bank - Remote

September 2025 - February 2026

Sr. Analyst - Servicing Analytics

- Delivered analytics, reporting, and strategic insights across Servicing, Collections, Capital Markets, Legal, and Compliance to monitor portfolio health and operations
- Analyzed repayment behavior and servicing trends; designed A/B tests and presented recommendations to leadership, clients, investors, and financial institutions

Solar Mosaic LLC - Remote/Oakland, CA

June 2021 - September 2025

Sr. Risk Analyst

- Developed SQL/Gephi network analysis using IP, geographic, and business-registration data to identify risk and fraud among connected entities
- Built an XGBoost dealer offboarding model to predict each dealer's probability of becoming offboarded at different time horizons (3M, 6M, 12M)
- Implemented dynamic partner risk scorecards, daily-updating schemas, alerts, and data initiatives with Data Engineering using DBT, Apache Airflow, AWS, Snowflake
- Monitored risk KPIs, conducted loan-level performance analysis, and maintained feedback loops with credit policy and operations to improve portfolio credit quality

L-Egant Solutions, LLC - Irvine, CA

October 2019 - June 2021

Lead Software Developer [Contract]

- Developed scripts and GUI software for high-frame-rate camera operation, video capture, image algebra, pixel-intensity analysis, and two-dimensional Gaussian characterization

Mechanics Bank Auto Finance - Irvine, CA

August 2019 - June 2021

Data Analyst - Portfolio & Servicing Risk

- Applied K-means, hierarchical clustering, and Random Forest models to segment collection agents, analyze credit metrics, inform credit policy, and forecast portfolio losses

Publications

Journal Articles

- [1] C. T. Chavez et al. "Measurement of 2D density profiles using a second-harmonic, dispersion interferometer." In: *Review of Scientific Instruments* 94.2 (Feb. 2023), p. 023503. ISSN: 0034-6748. URL: <https://doi.org/10.1063/5.0119896>.
- [2] M. Nikolic et al. "Applicability of optical emission spectroscopy techniques for characterization of Ar and Ar/O₂ discharges." In: *Journal of Physics D: Applied Physics* 54.27 (Apr. 2021), p. 275203. URL: <https://dx.doi.org/10.1088/1361-6463/abf61c>.

Skills

- Version Control (GitHub, Bitbucket)
- DBT, AirFlow, Docker, AWS, Snowflake
- Python, SQL, R, LaTeX, Java
- Network Analysis & Gephi
- Matplotlib, Mayavi, ggplot2, Tableau
- Monte Carlo Simulations